

Issue # 7: Numerator Data - Trends and Event Studies 2023-2024

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This Write-up contains tables and plots for trend and event study analyses of the Numerator data. The original user-trip-item level data is transformed to user- month level data, with variables summarized at the monthly level for each user. SNAP and Non-SNAP treatment is determined by SNAP participation rate of each user over the periods of the data (>0.5 on SNAP; ≤ 0.5 Non-SNAP); Not adjusted for inflation

1. Figures: Monthly average of ...

- level/log total spending winsorized
- level/log Food spending winsorized
- level/log Food on EBT winsorized
- level/log Soda spending winsorized
- level/log Soda on EBT winsorized
- prop. food spending
- prop. food on EBT
- prop. soda
- prop. soda on EBT
- level/log soda spending by SNAP/Non-SNAP
- level/log soda spending by state $\times$ SNAP

2. Figures: Event Studies

- level/log soda spending winsorized
- level/log soda spending by SNAP $\times$ post (conditional on treated states)
- level/log soda spending DDD

3. Tables

- TWFE/DDD Model Estimates

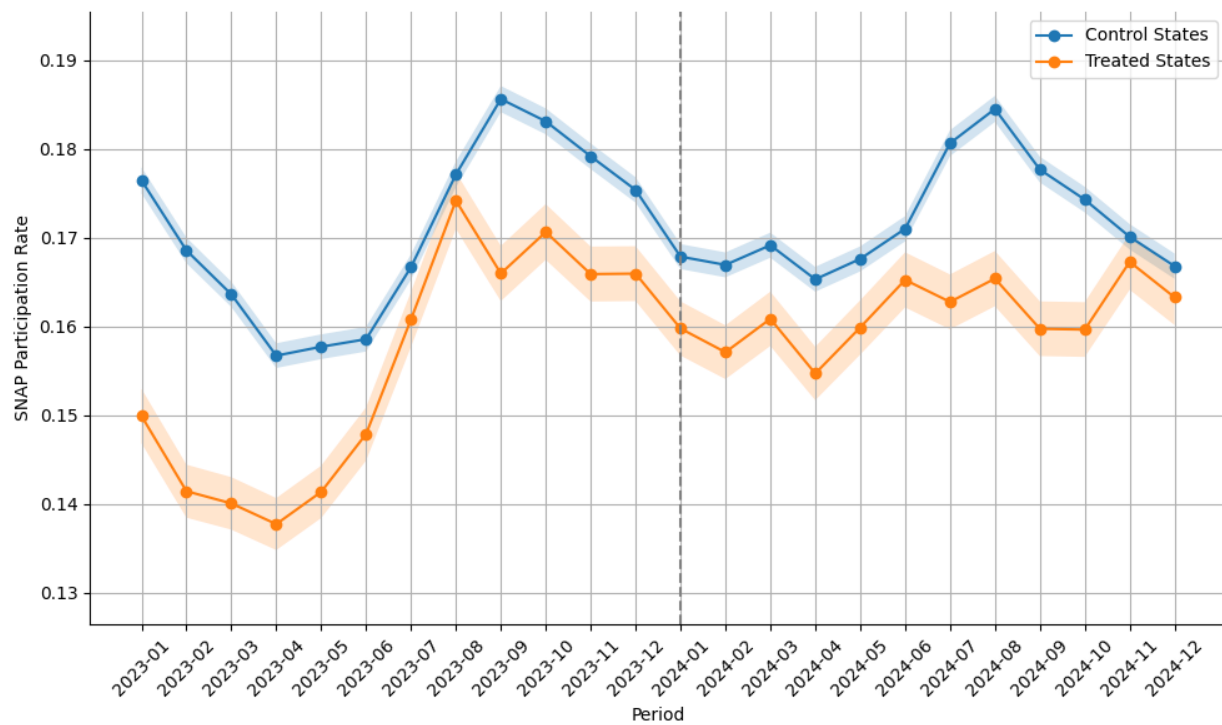
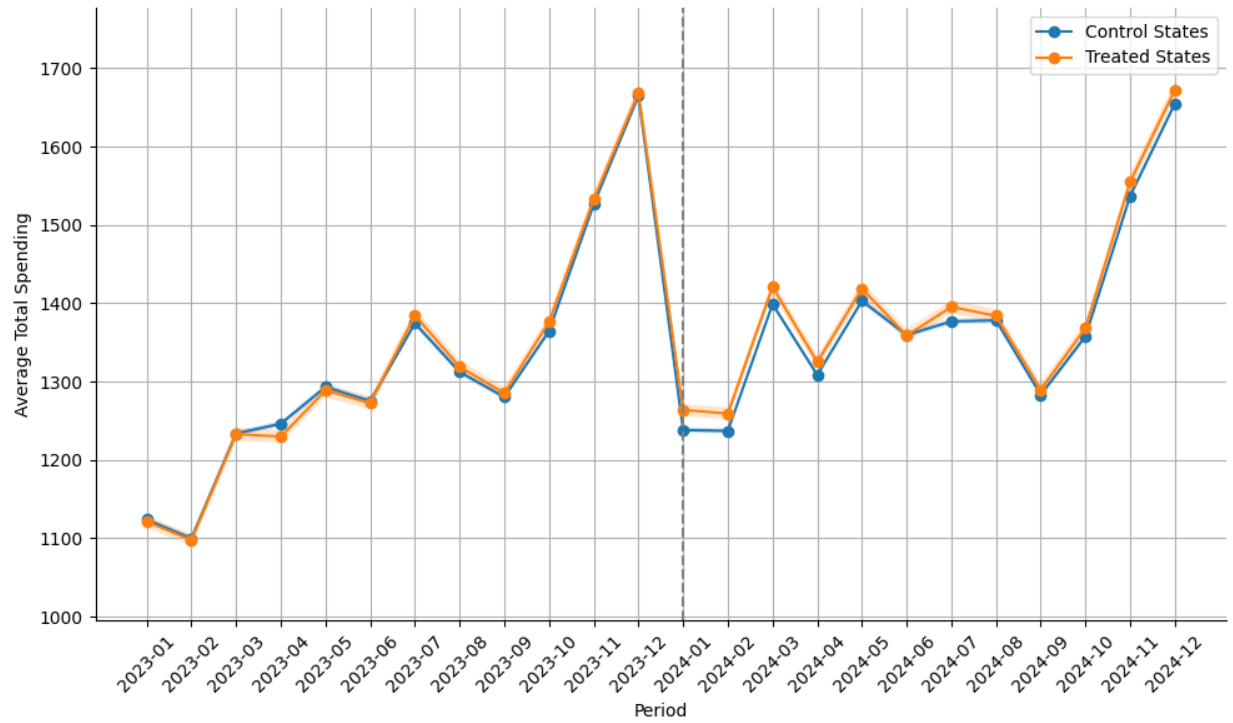


Figure 1: SNAP Participation Rate

Figure 2: Monthly Average of Total Spending

(a) level



(b) log

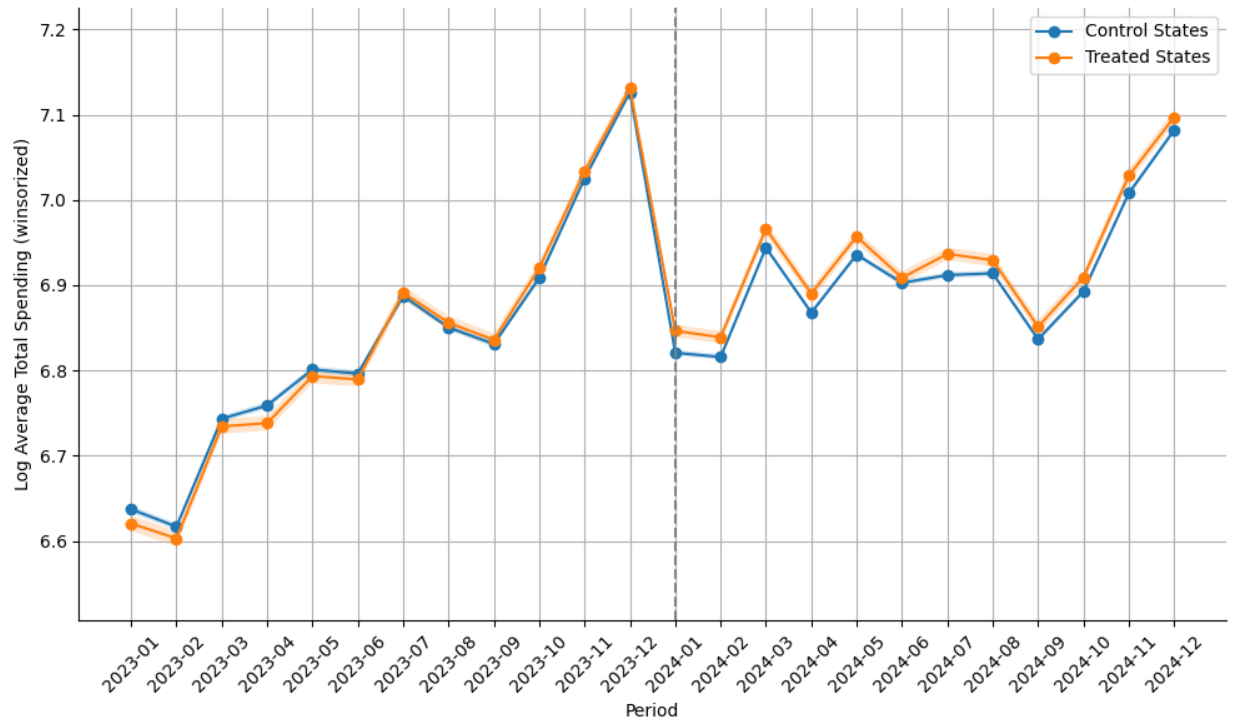
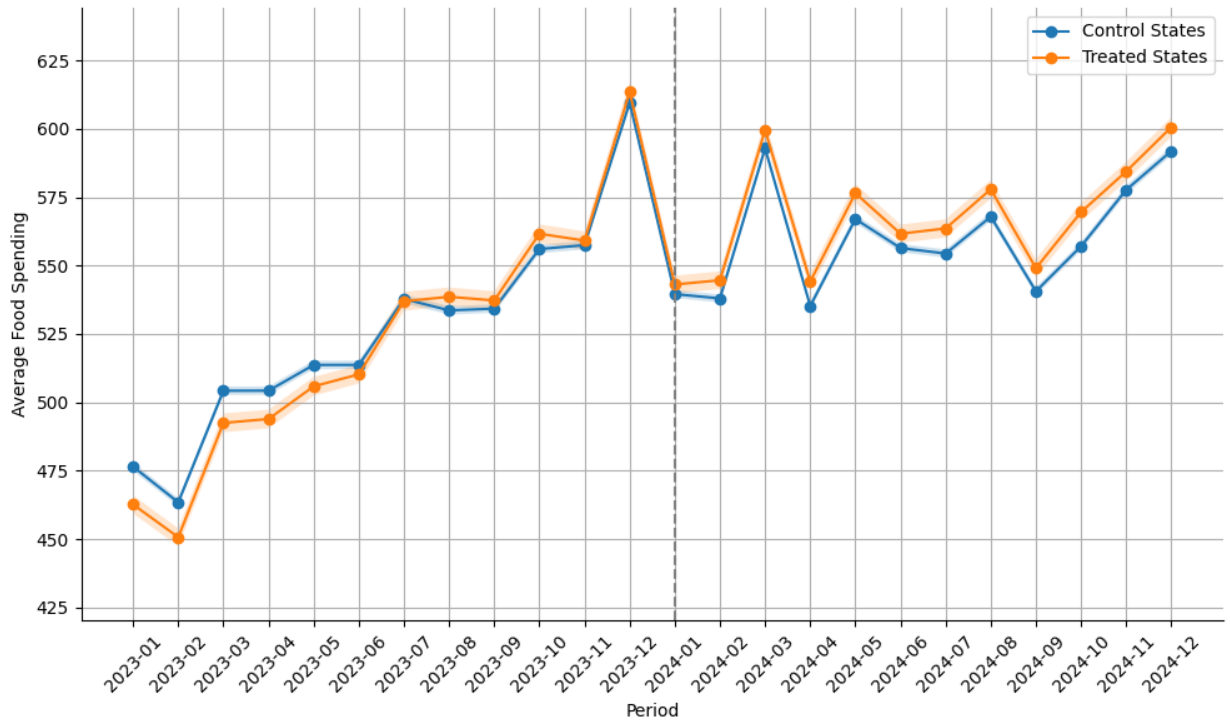


Figure 3: Monthly Average of Food Spending

(a) level



(b) log

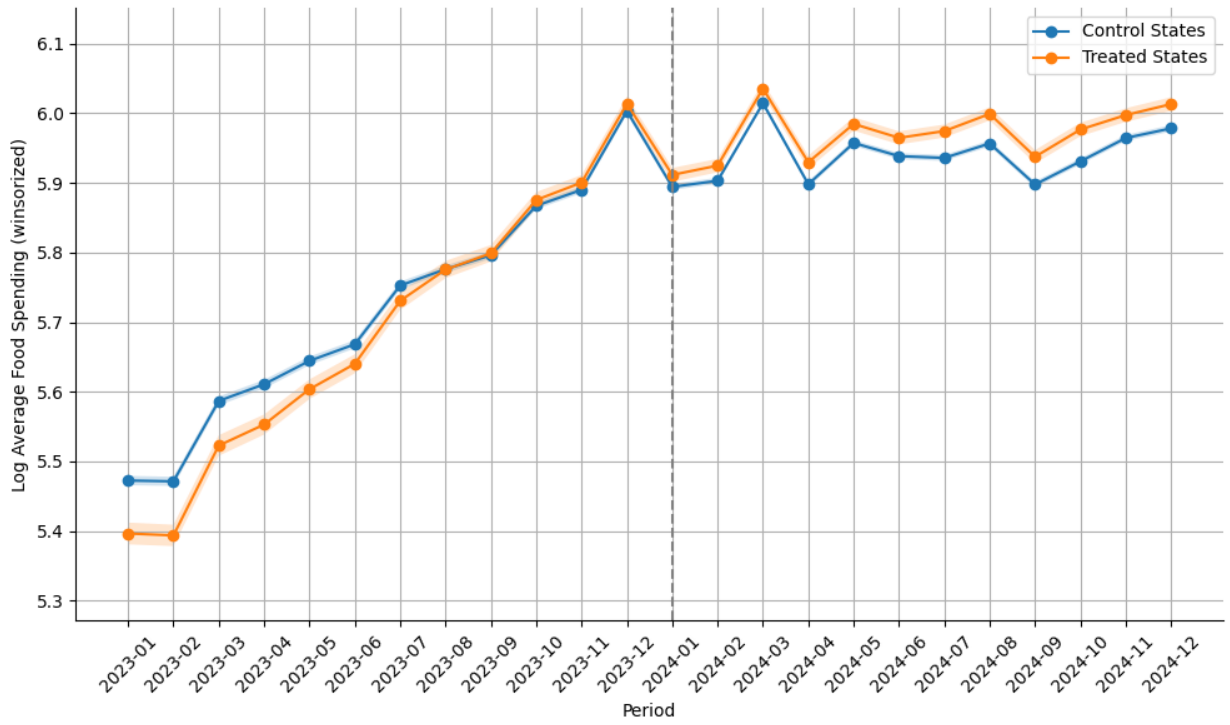
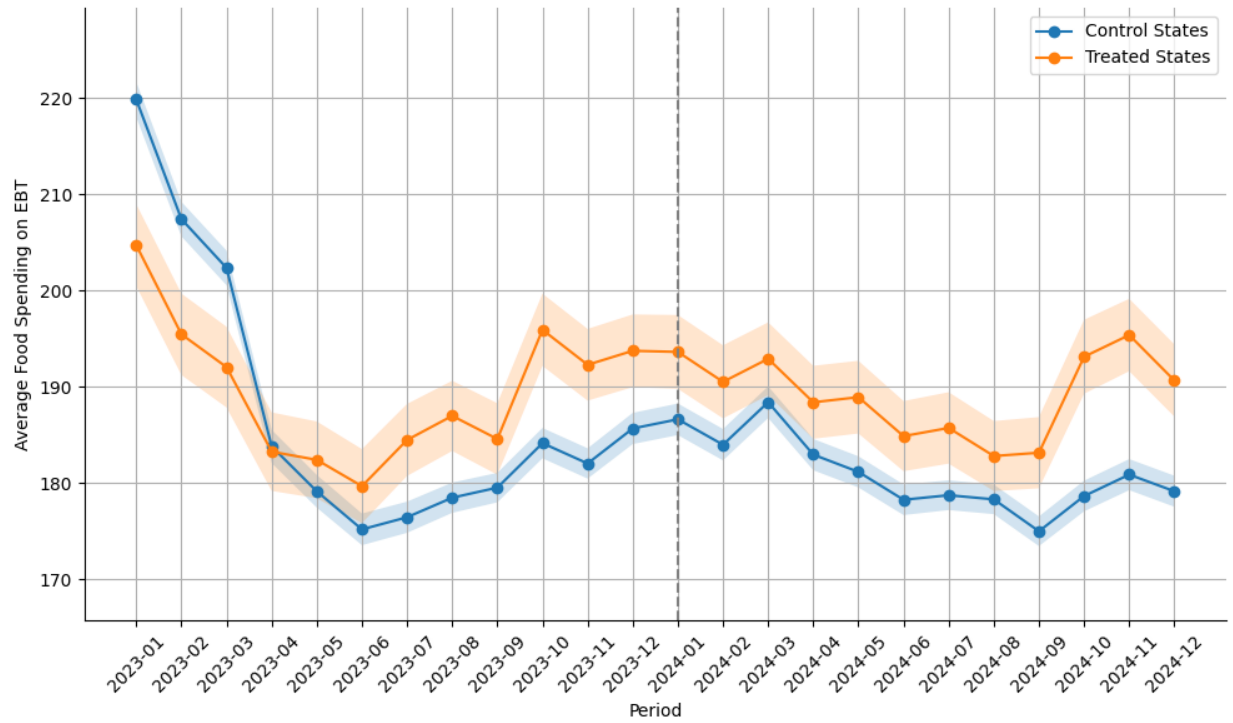


Figure 4: Monthly Average of Food Spending on EBT

(a) level



(b) log

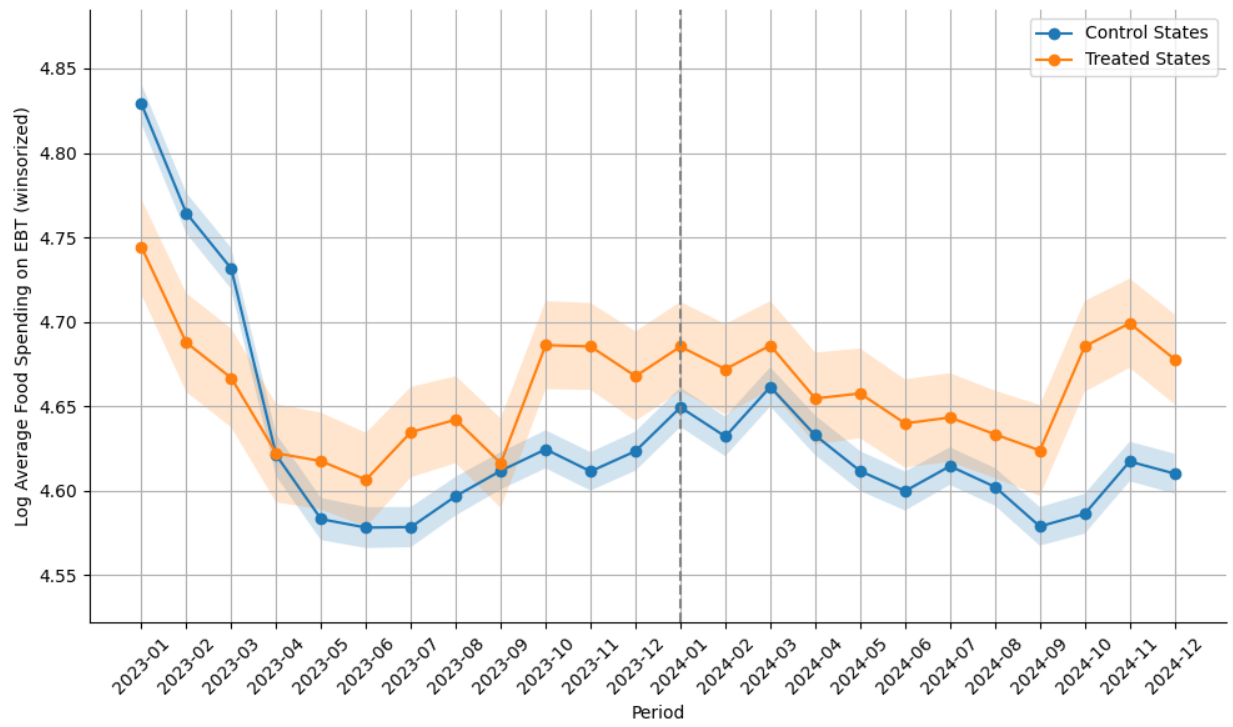
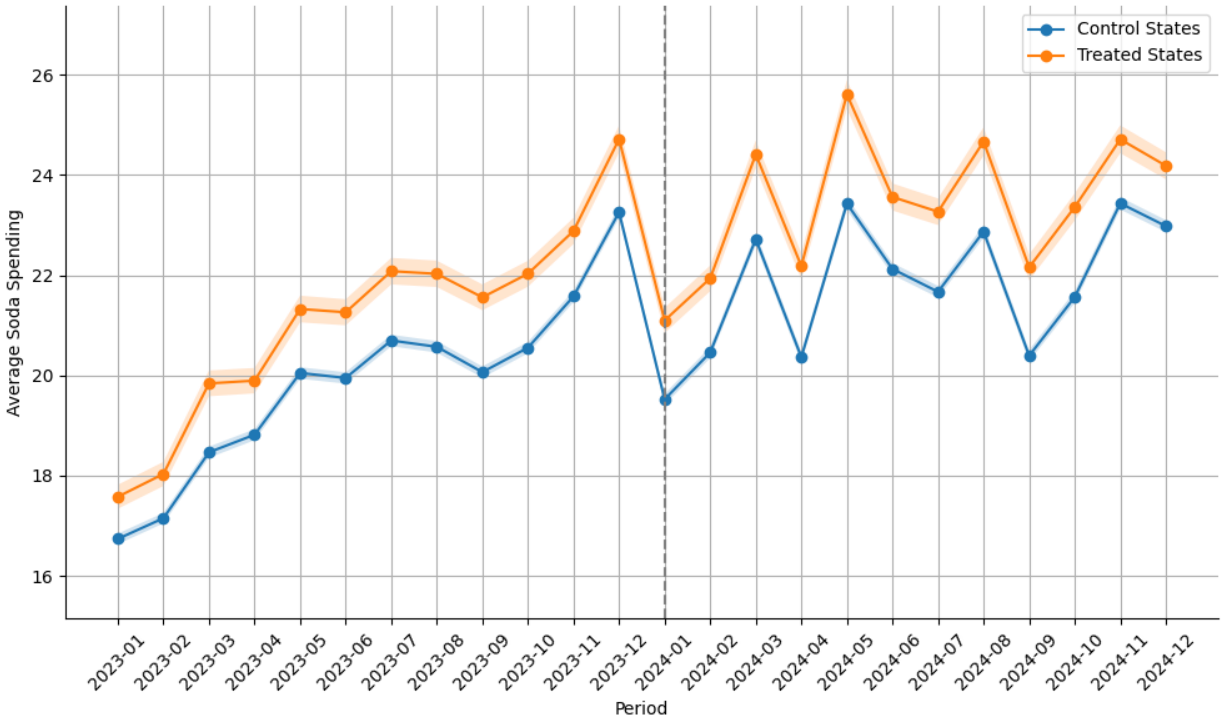


Figure 5: Monthly Average of Soda Spending

(a) level



(b) log

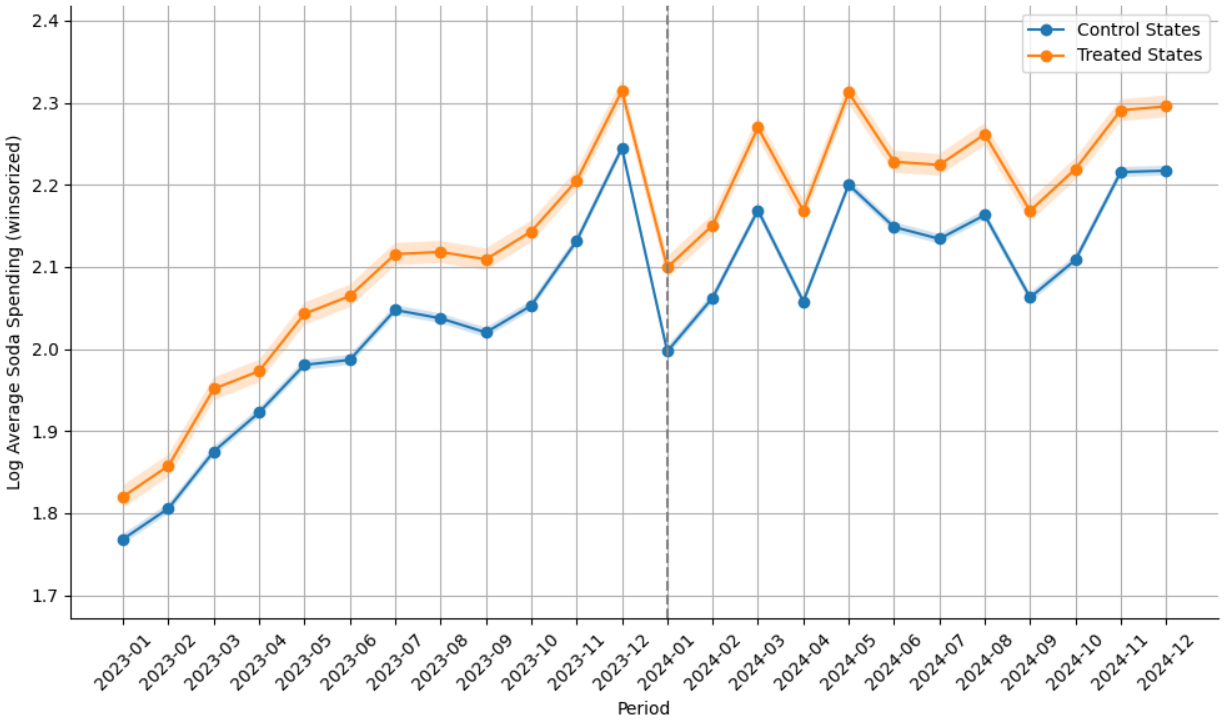
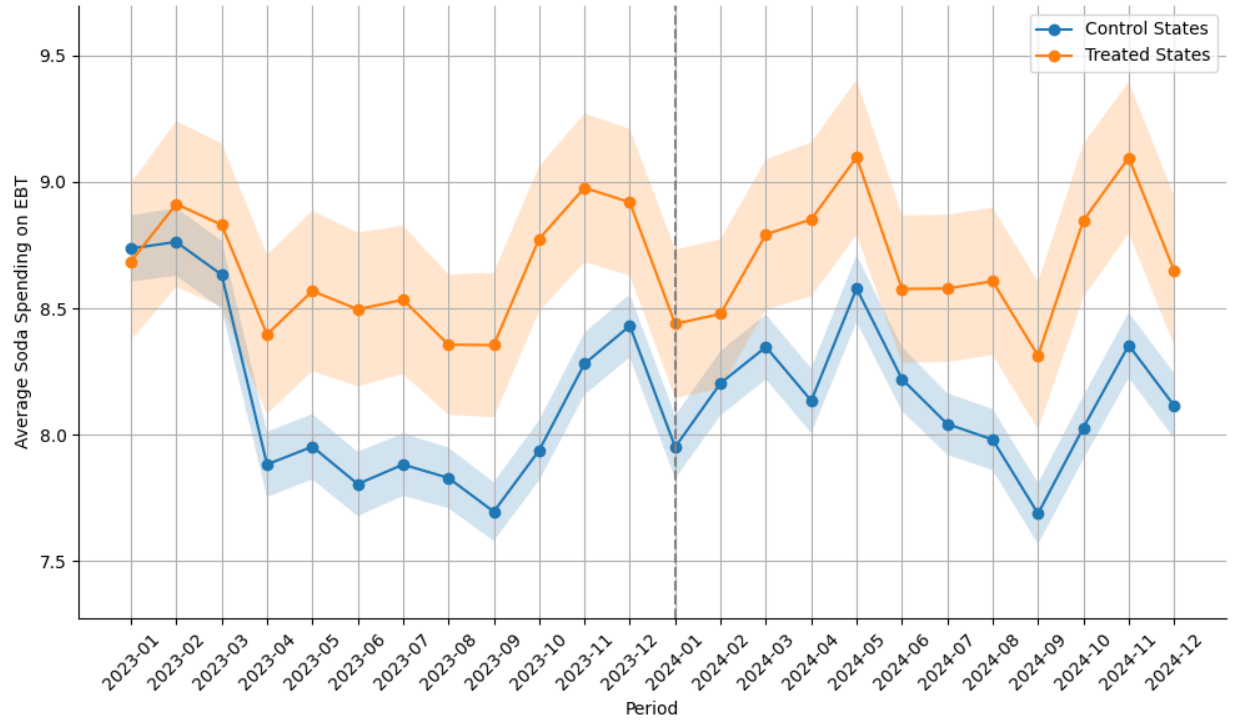


Figure 6: Monthly Average of Soda Spending on EBT

(a) level



(b) log

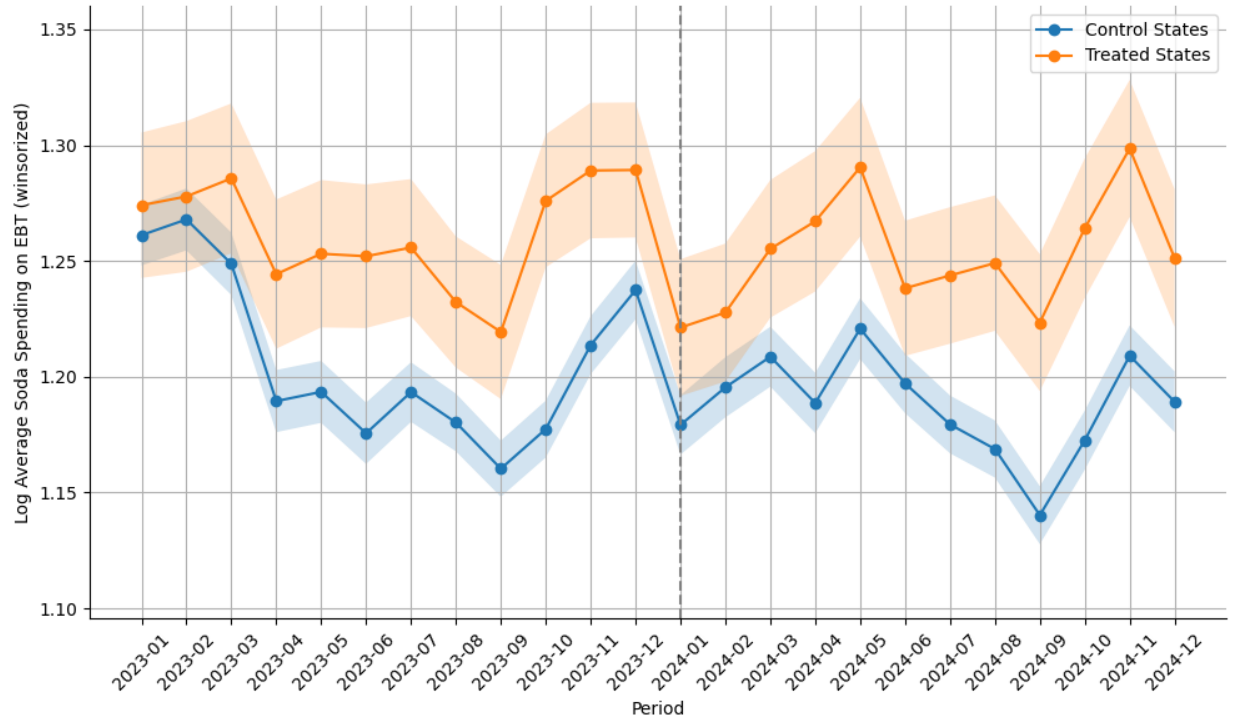


Figure 7: Monthly Average of Soda Spending by SNAP/Non-SNAP

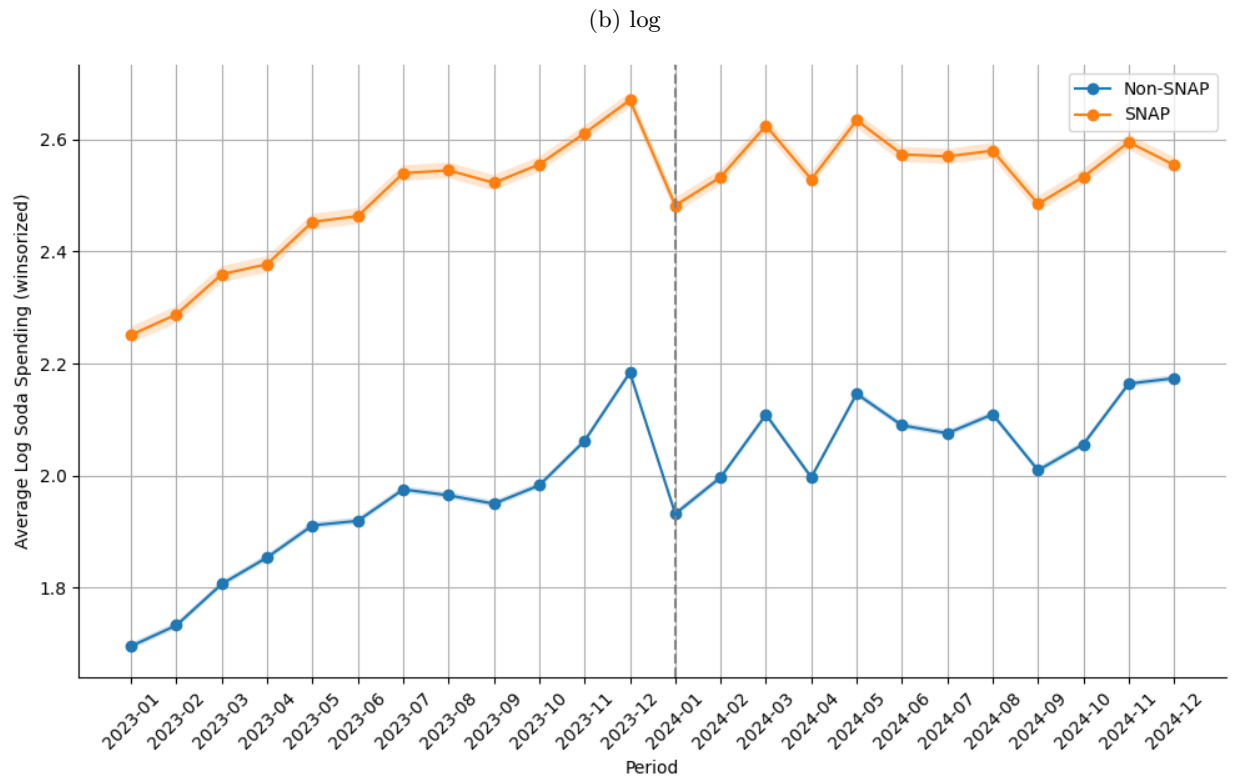
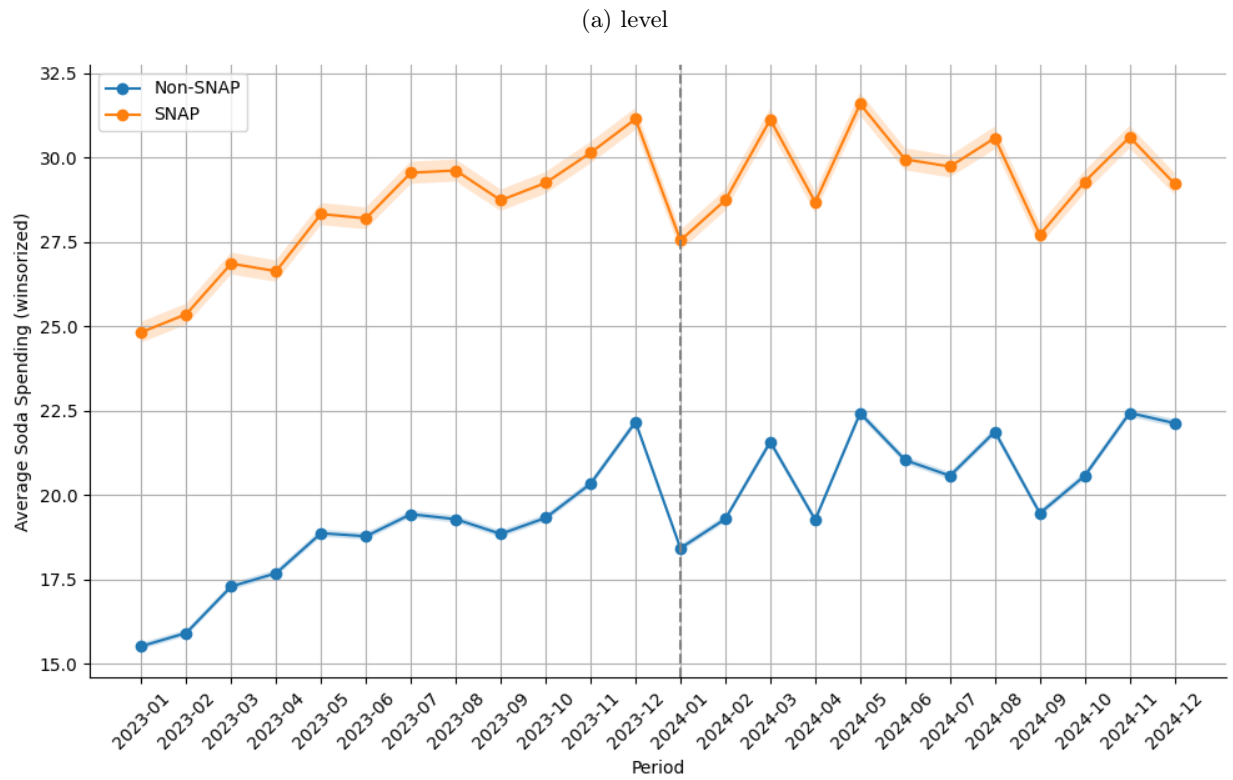
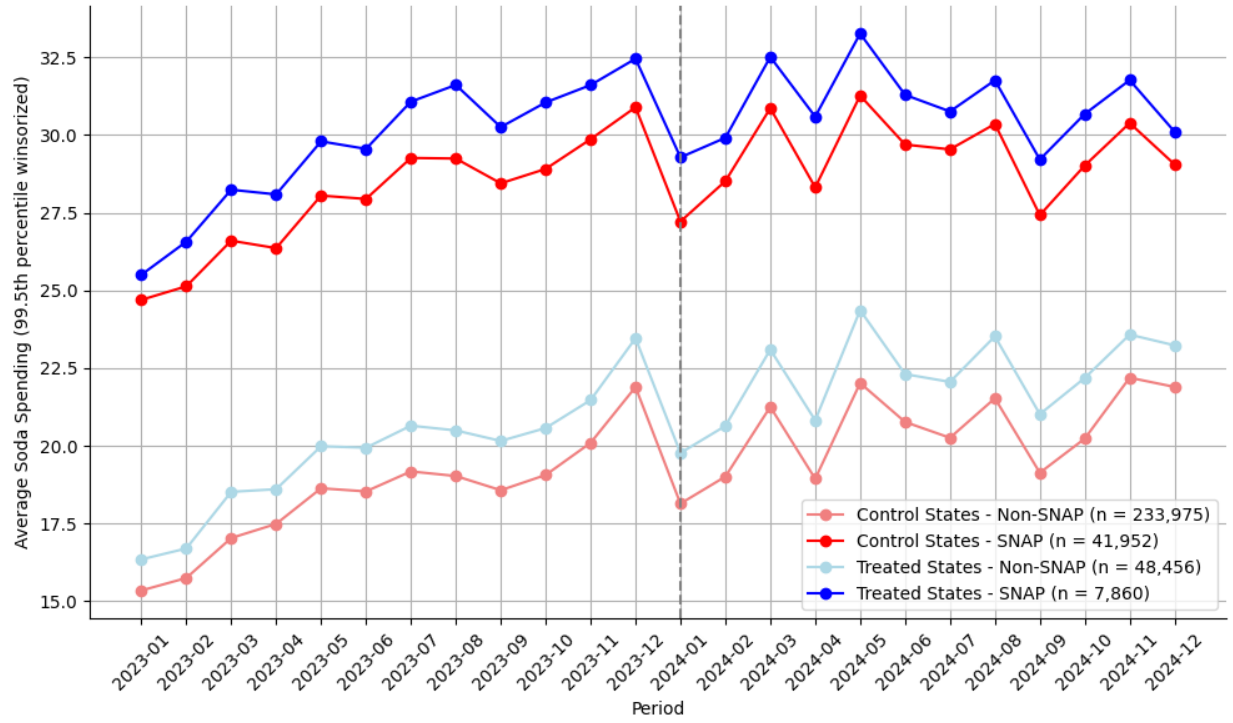


Figure 8: Monthly Average of Soda Spending by SNAP $\times$ state

(a) level



(b) log

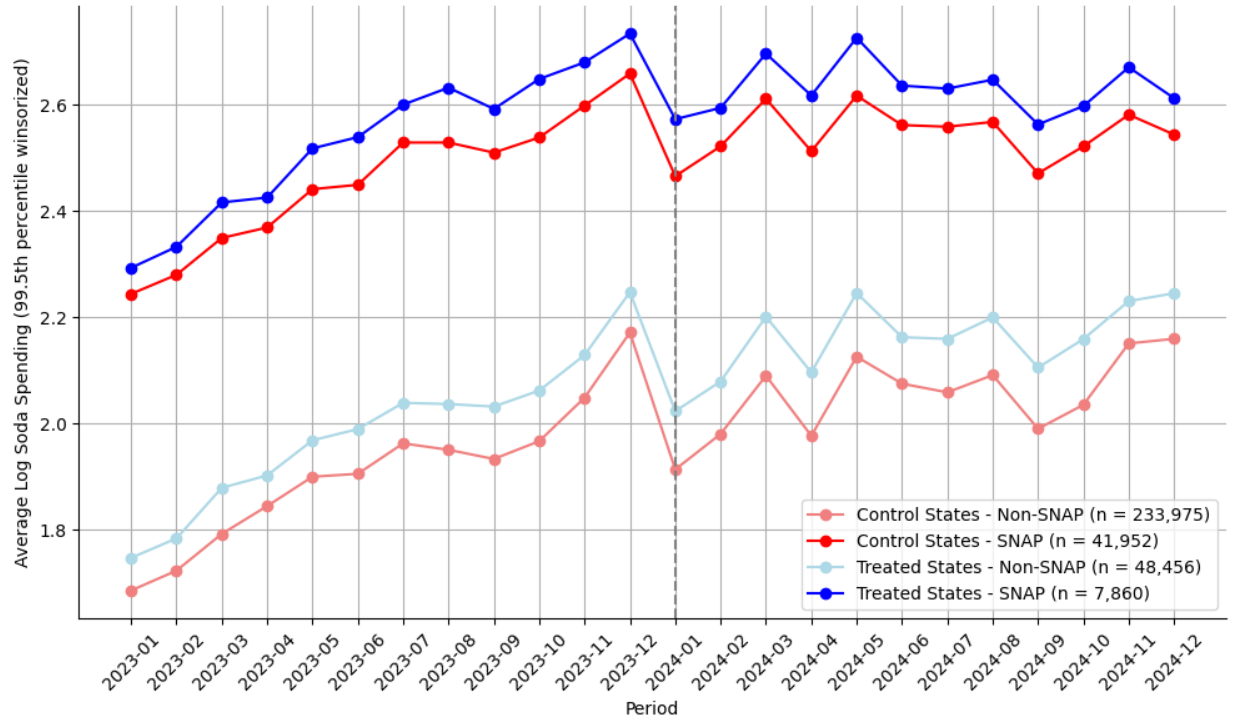


Figure 9: Proportion of Total Spending on Food

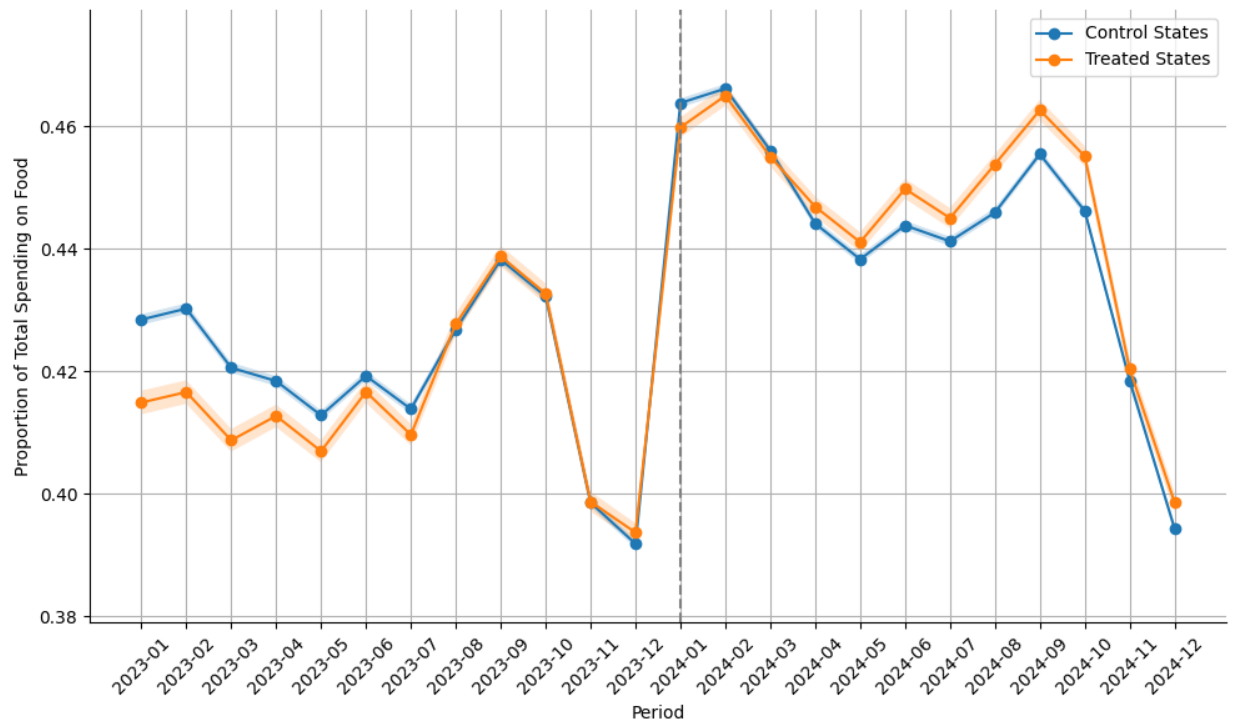


Figure 10: Proportion of Food Spending on EBT

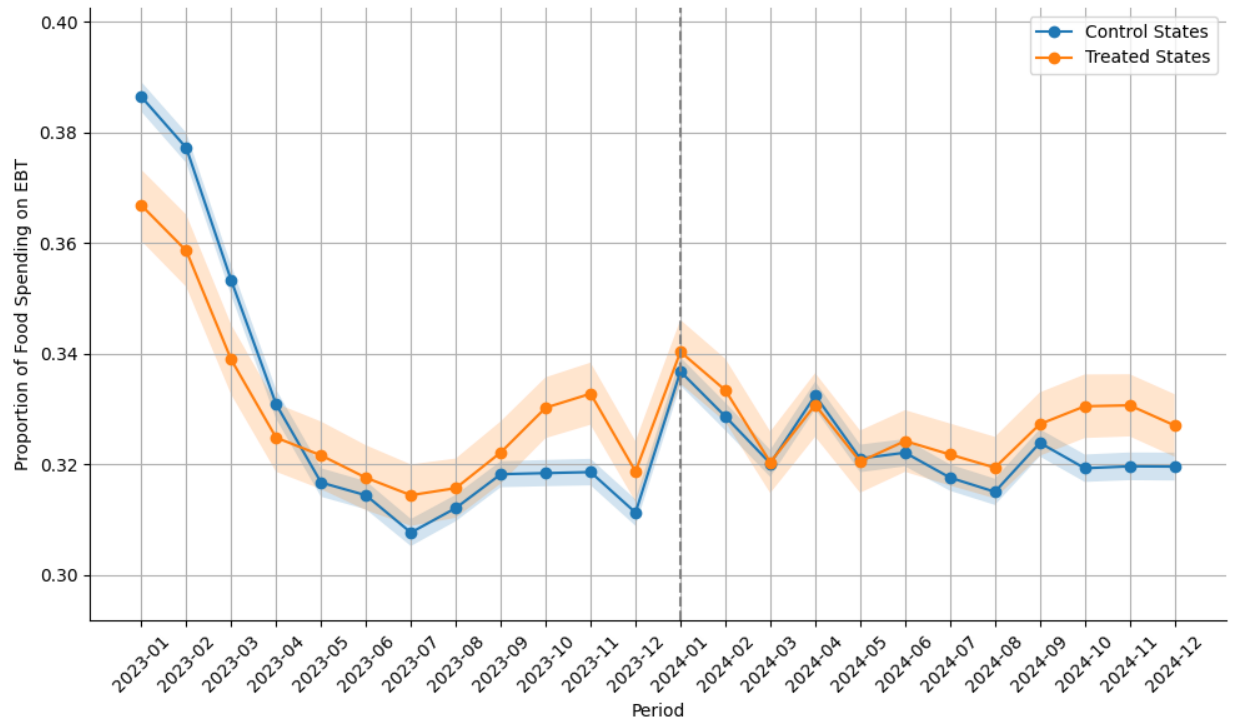


Figure 11: Proportion of Food Spending on Soda

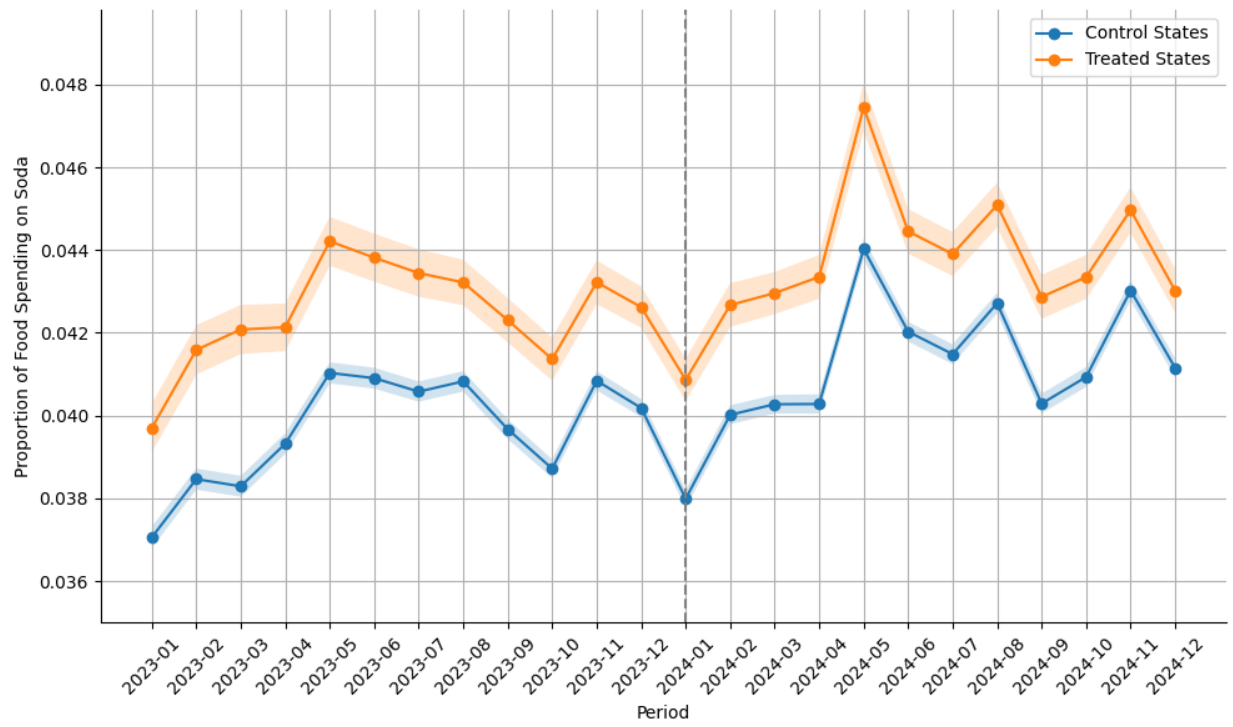


Figure 12: Proportion of EBT Spending on Soda

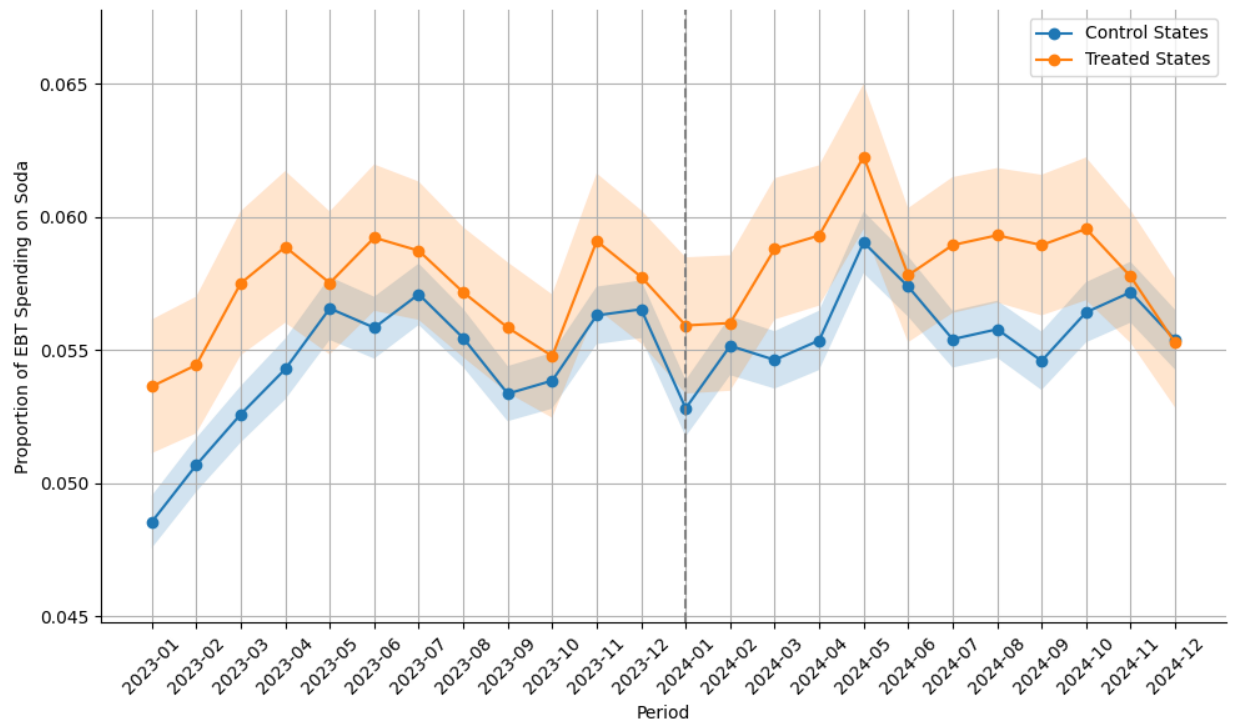


Figure 13: Proportion of Soda Spending on EBT

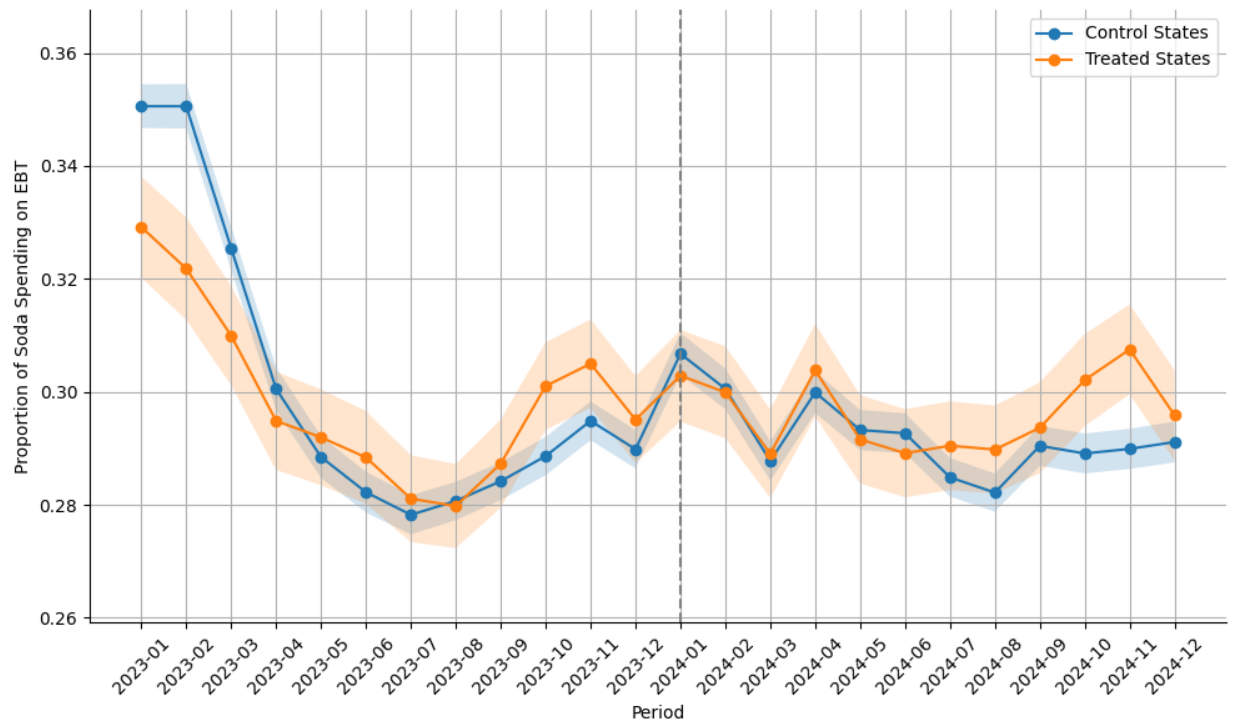
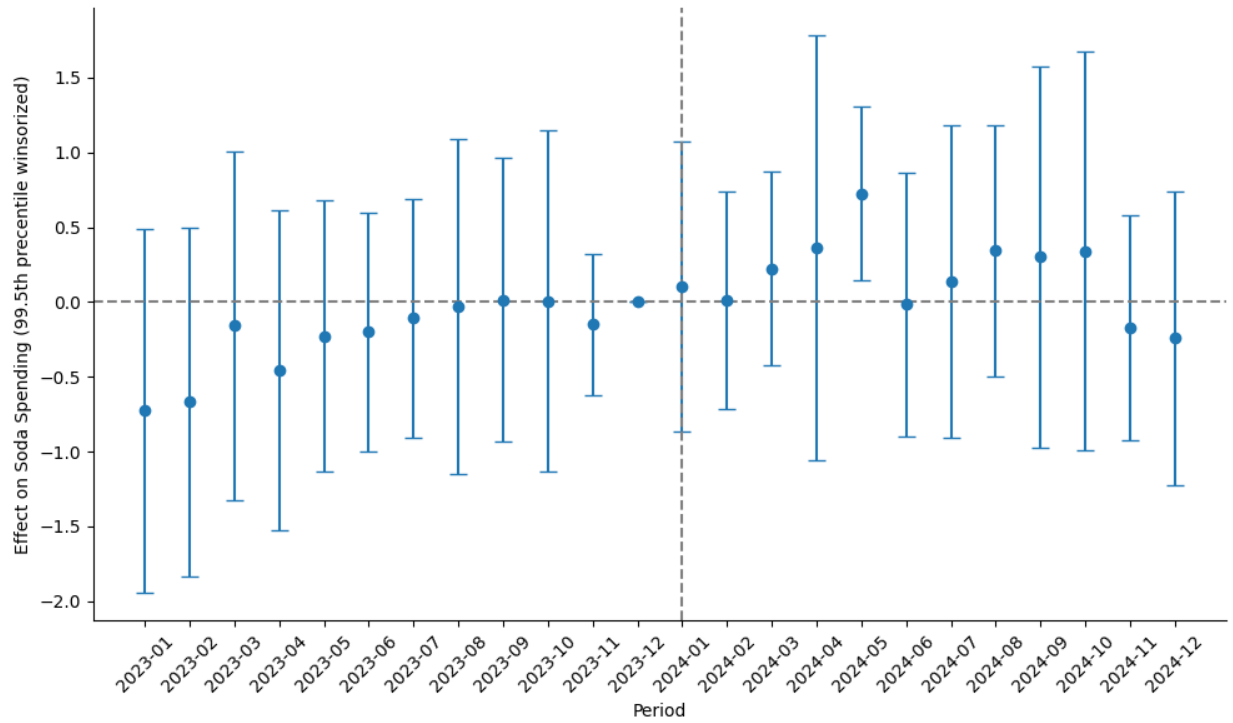


Figure 14: Event Study - TWFE

(a) level



(b) log

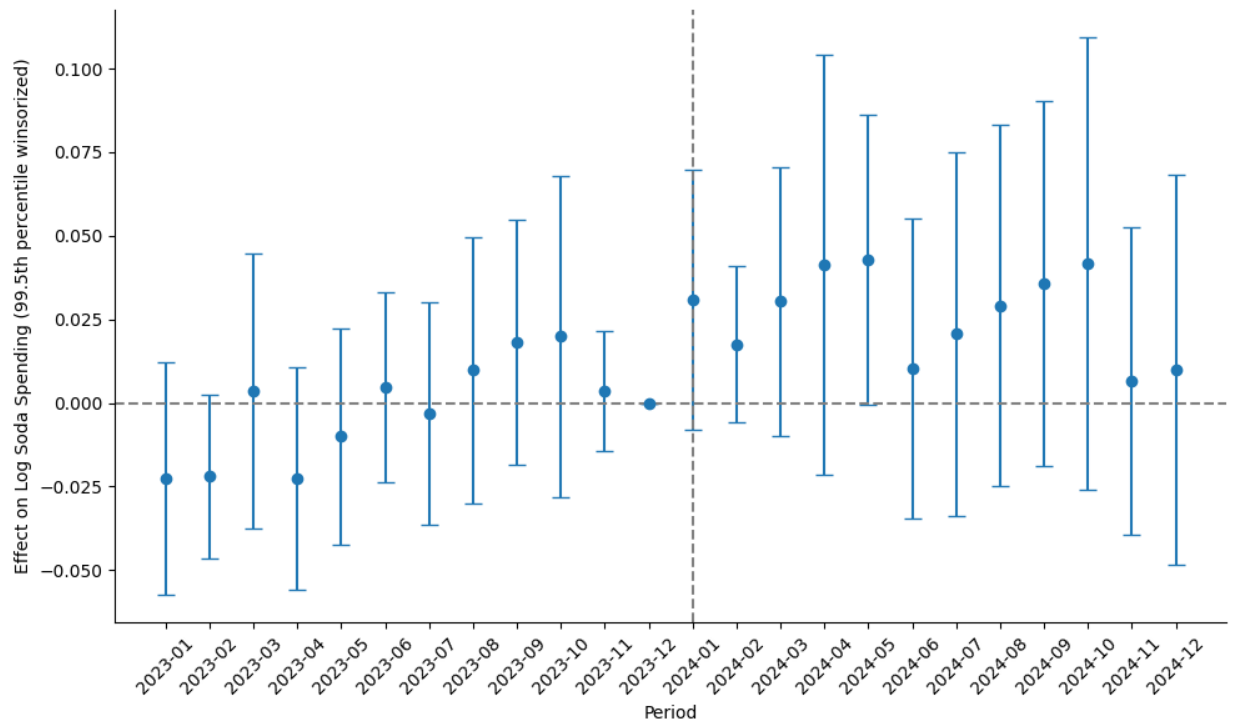


Figure 15: Event Study - TWFE (Treatment Assignment = SNAP/Non-SNAP) within treated states

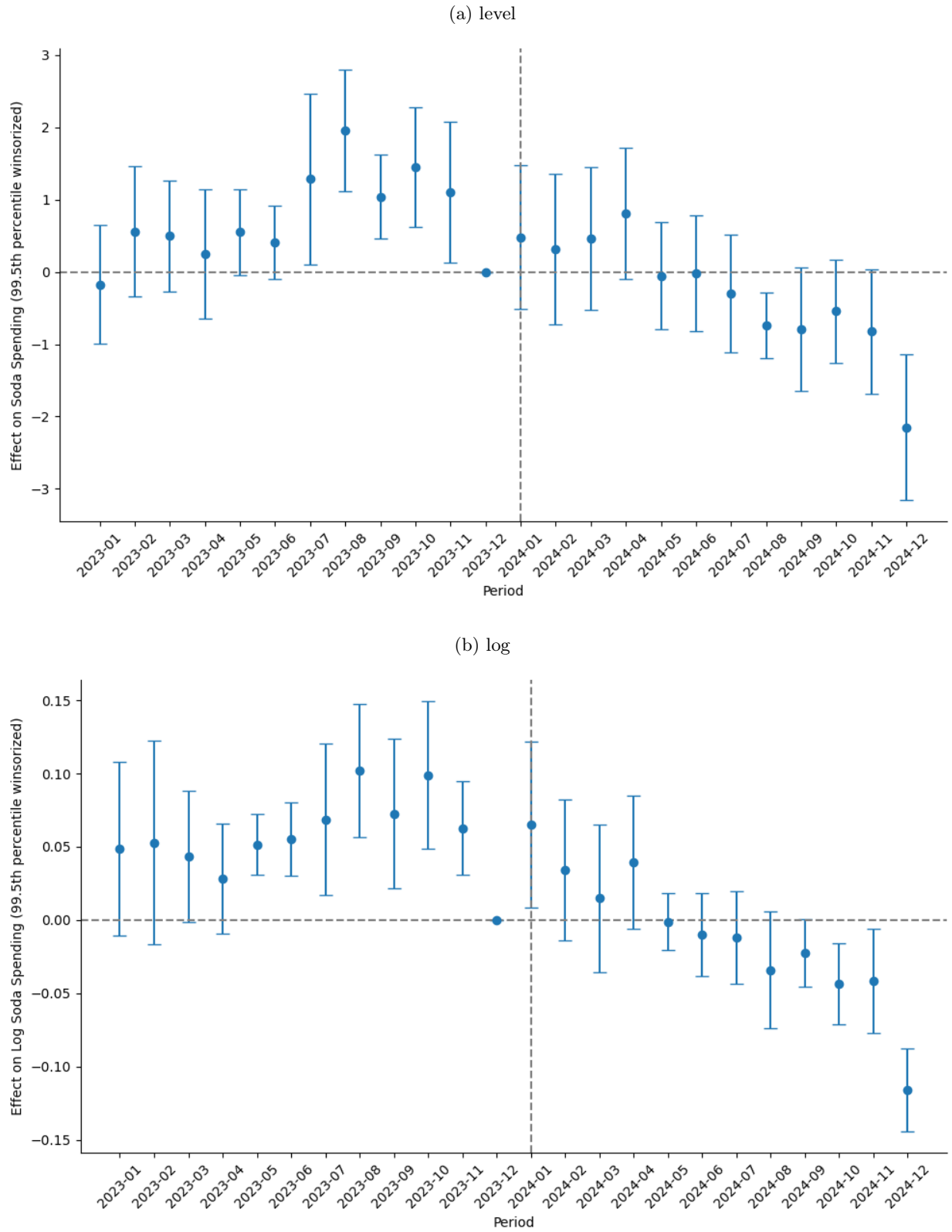


Figure 16: Event Study - DDD

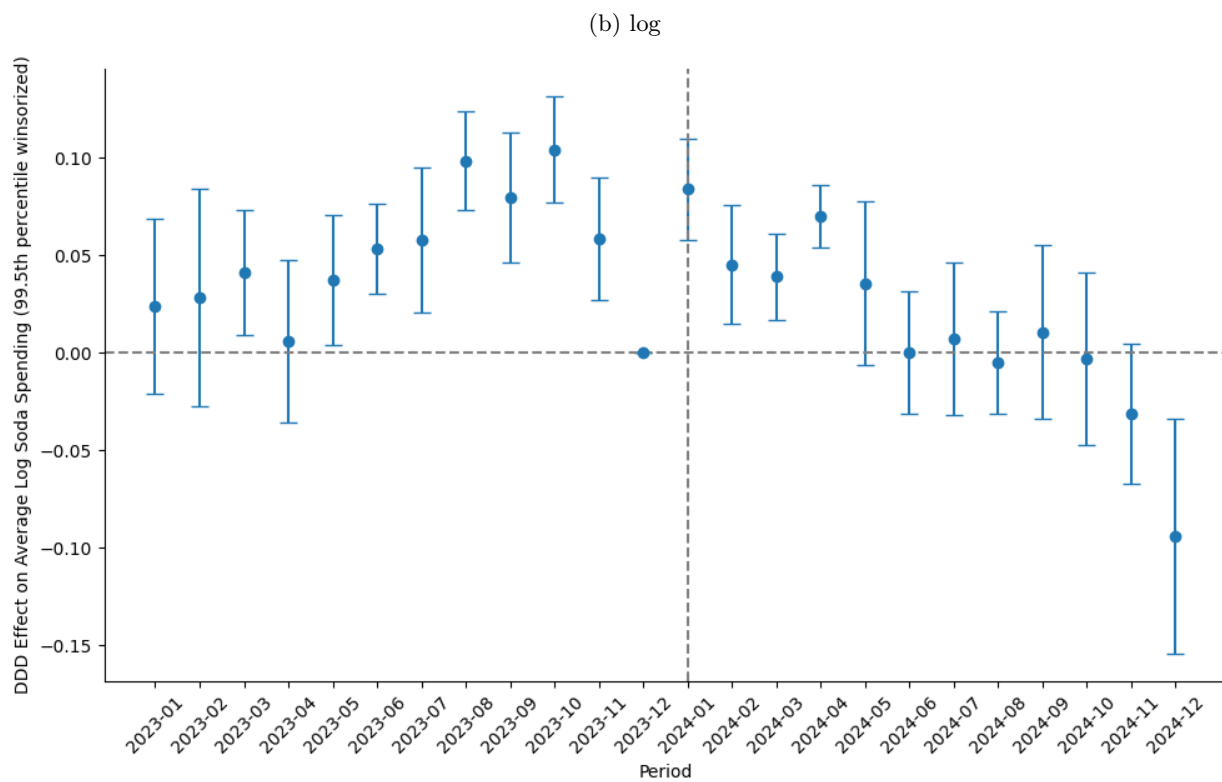
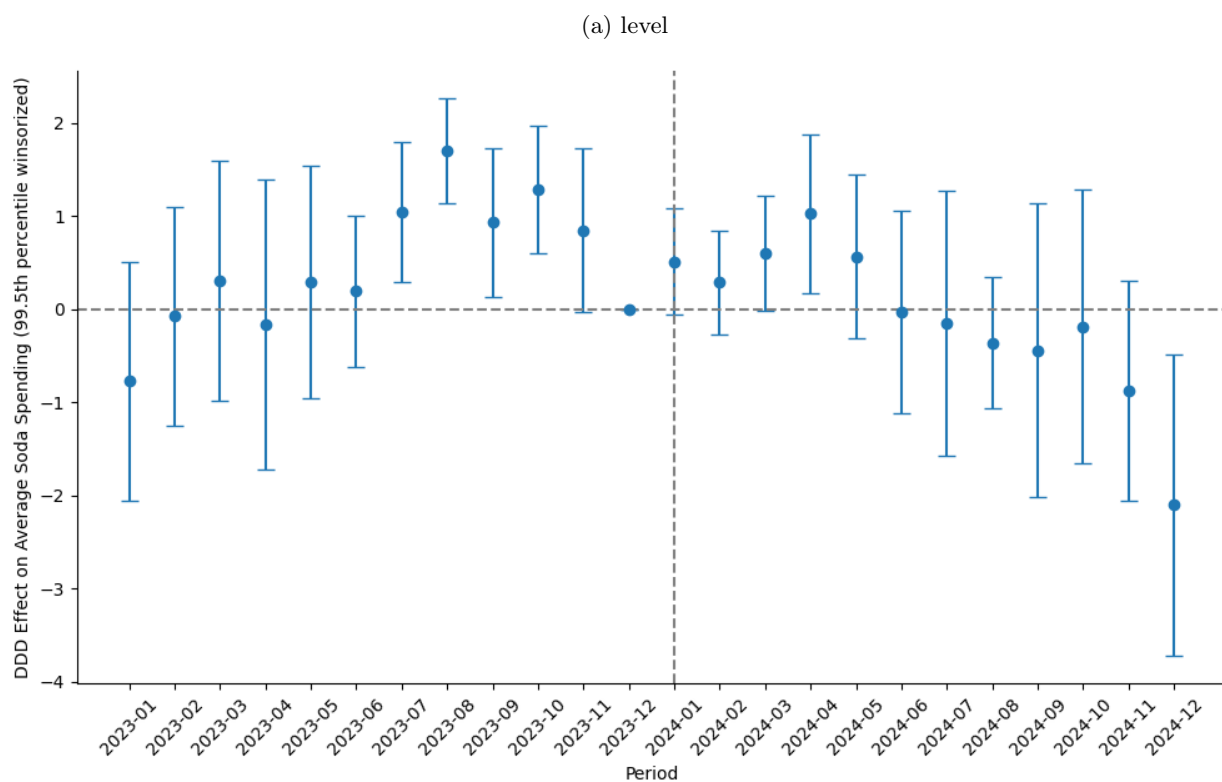


Table 1: Model Estimates

Term	TWFE	TWFE Log	TWFE	TWFE Log	DDD	DDD Log
Post	1.9256*** (0.0988)	0.1343*** (0.0057)	2.4608*** (0.0878)	0.1713*** (0.0130)	2.0055*** (0.0979)	0.1398*** (0.0064)
State $\times$ Post	0.3908*** (0.1400)	0.0275** (0.0139)				
SNAP $\times$ Post			-1.0536*** (0.2135)	-0.0694*** (0.0085)		
State $\times$ SNAP $\times$ Post					-0.5983** (0.2394)	-0.0380*** (0.0127)
N	7,731,675	7,731,675	1,308,407	1,308,407	7,731,675	7,731,675
R^2	0.0026	0.0040	0.0035	0.0055	0.0026	0.0040
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
SE Cluster	state	state	state	state	state	state